

SK SUMIT AHMED

Kolkata, West Bengal, India

+91 980414XXXX ✉ sksumitahmed007@gmail.com 🐦 @sumitkun_
🌐 linkedin.com/in/sumitxahmed 🏠 github.com/sumitahmed 🌐 sumitahmed.me

Technical Skills

Languages: TypeScript, JavaScript, Java, C, C++, SQL, Python

Web & Backend: React, Next.js, Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, WebRTC

Databases & Tools: PostgreSQL, Prisma, MongoDB, MySQL, Supabase, Docker, Git, Linux, Cloudflare, Google Cloud Platform

AI & Computer Vision: PyTorch, TensorFlow, OpenCV, Computer Vision, Image Processing, RAG, Vector Databases

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Experience

Research Intern — Jadavpur University

May 2026 – Present

Department of Electronics & Telecommunication Engineering

Kolkata, India

- Conduct research on image and video dehazing using deep learning and computer vision for image restoration.
- Co-author of the survey paper *A Review of Dehazing Benchmark Datasets*; contribute to ongoing video-dehazing research.
- Develop and evaluate Python, OpenCV, PyTorch, and TensorFlow pipelines using benchmark datasets, PSNR, and SSIM.

Freelance Web Developer

2024 – Present

- Delivered **5 client projects**, including developer portfolios, a coaching-center student management system, blog pages, and web applications.
- Worked end-to-end from client requirements and UI implementation to backend integration, databases, testing, and production deployment.

Open Source Contributor

2025 – Present

- Continuously contribute to open-source projects to gain real-world development experience, work with maintainers, learn through reviews and discussions, and understand large production codebases; **5+ contributions merged** so far.

Projects

Panvas – Local-First Visual Research Workspace

🔗 Live | 🏠 GitHub

React, TypeScript, Electron, Vite, Excalidraw, IndexedDB, Google Drive API

- Built a local-first Windows and web workspace combining structured notebooks, vector handwriting, rich-text notes, PDF annotation, and an infinite visual canvas.
- Designed separate desktop filesystem and browser IndexedDB persistence layers, with search, backup/restore, media notes, PDF export, and optional Google Drive sync.

FreeCompute – Remote GPU AI Development Platform

🔗 Live | 🏠 GitHub

Python, Kaggle GPUs, Cloudflare Tunnel, LLM Inference, CLI Tooling

- Built a local development harness that runs open-source LLM and image-model inference on free remote GPUs while keeping source code and file operations local.
- Connected Kaggle dual-T4 sessions to local workflows with authenticated inference, secure tunneling, approval-aware tools, reversible edits, journaling, and automated tests.

TabBridge – Cross-Browser Tab Group Transfer

🏠 GitHub

JavaScript, Chrome Extension APIs, Manifest V3

- Built a zero-backend Chromium extension to export and restore tab groups across Chrome, Brave, Edge, and other Chromium browsers.
- Preserved tab URLs, group names, colors, and organization through a portable JSON format with all processing performed locally.

Certifications and Achievements

- **Oracle Cloud Infrastructure 2025 Certified Networking Professional** — Certificate
- **Oracle Agentic AI Certified Foundations Associate** — Certificate
- **Java Programming Certificate** — GeeksforGeeks
- Solved **400+ Data Structures and Algorithms problems** across LeetCode and GeeksforGeeks.

Education

RCC Institute of Information Technology

2023 – 2027

B.Tech in Electronics and Communication Engineering

Kolkata, India